

Exercises: Gradient algorithms

Exercise 1 (A quadratic example in $\mathbb{R}^2$). Consider, for $\gamma > 0$, $f(x) = \frac{1}{2}(x_1^2 + \gamma x_2^2)$. We apply the gradient descent method with optimal step, starting at $x^{(0)} = (\gamma, 1)$.

1. Show that f is m -convex with M -Lipschitz gradient. Find the tightest m and M constants.
2. Show that

$$x^{(k)} = \left(\gamma \left(\frac{\gamma-1}{\gamma+1} \right)^k, \left(-\frac{\gamma-1}{\gamma+1} \right)^k \right)$$

and

$$f(x^{(k)}) = \frac{\gamma(\gamma+1)}{2} \left(\frac{\gamma-1}{\gamma+1} \right)^{2k} f(x^{(0)})$$

3. Show that, on this example, the convergence is exactly linear, that is $f(x^{(k)}) - v^\#$ is a geometric series. Give its reason. Compare with the theoretical bound.
4. When is this algorithm fast and slow ?

Exercise 2 (Strongly convex - optimal step). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a m -convex $\mathcal{C}^2$ function. Define, for given $x^{(0)}$,

$$\begin{aligned} \tilde{f}_k : t &\mapsto f(x^{(k)} - t \nabla f(x^{(k)})) \\ t^{(k)} &= \arg \min_{t \in \mathbb{R}} \tilde{f}_k(t) \\ x^{(k+1)} &= x^{(k)} - t^{(k)} \nabla f(x^{(k)}) \end{aligned}$$

1. Show that there exists $M \geq m$ such that $mI \preceq \nabla^2 f(x^{(k)}) \preceq MI$
2. Show that, for any interesting t (to be defined) we have

$$\tilde{f}_k(t) \leq f(x^{(k)}) - t \|\nabla f(x^{(k)})\|_2^2 + \frac{Mt^2}{2} \|\nabla f(x^{(k)})\|_2^2$$

3. Show that,

$$f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{1}{2M} \|\nabla f(x^{(k)})\|_2^2$$

4. Show that $f(x^\#) \geq f(x) - \frac{1}{2m} \|\nabla f(x)\|_2^2$
5. Show that

$$f(x^{(k+1)}) - f(x^\#) \leq \left(1 - \frac{m}{M}\right) [f(x^{(k)}) - f(x^\#)]$$

6. Show that the algorithm converges, and give its convergence speed.

Answers:

1. Let $S = \text{lev}_{f(x^{(0)})} f$. By strong convexity it is a bounded set. f being $\mathcal{C}^2$, its Hessian is continuous and thus bounded on S , where all $x^{(k)}$ lives.
2. For any t such that $y := x^{(k)} - t \nabla f(x^{(k)}) \in S$, there exists $z \in [x^{(k)}, y]$, such that

$$\begin{aligned} f(y) &= f(x^{(k)}) - \nabla f(x^{(k)})^\top (y - x^{(k)}) \\ &\quad + \frac{1}{2} (y - x^{(k)})^\top \nabla^2 f(z) (y - x^{(k)}) \end{aligned}$$

replacing y by its value, and using the upper bound on $\nabla^2 f(z)$ yields the result.

3. Use $t = 1/M$ in the upperbound of the previous question. Note that this choice of t minimizes said upper bound.
4. We have, by strong convexity,

$$\begin{aligned} f(y) &\geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|^2 \\ &\geq f(x) + \nabla f(x)^\top (\tilde{y} - x) + \frac{m}{2} \|\tilde{y} - x\|^2 \\ &= f(x) - \frac{1}{2m} \|\nabla f(x)\|^2 \end{aligned}$$

where $\tilde{y}$ minimizes the right hand side. We then apply to $y = x^\#$.

5. By the previous question we have $\|\nabla f(x^{(k)})\|^2 \geq 2m(f(x^{(k)}) - v^\#)$. Question 3 then yields

$$f(x^{(k+1)}) \leq f(x^{(k)}) - m/M(f(x^{(k)}) - v^\#)$$

subtracting $v^\#$ on each sides yields the result.

6. Recursively we get $f(x^{(k)}) - v^\# \leq c^k(f(x^{(0)}) - v^\#)$, with $c = 1 - m/M$. In particular, for any $\varepsilon > 0$, we have $f(x^{(k)}) - v^\# \leq \varepsilon$ after at most $\frac{\ln(\varepsilon) - \ln(f(x^{(0)}) - v^\#)}{\ln(c)}$ iterations.

Exercise 3 (Strictly convex case). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^2 , convex function with M -Lipschitz gradient such that $f(x^\#) = \inf f$. We define, for given $x^{(0)}$.

$$x^{(k+1)} = x^{(k)} - t\nabla f(x^{(k)})$$

with $t \leq 1/M$.

1. Show that, for all x and y $(y-x)^\top \nabla^2 f(z)(y-x) \leq M\|y-x\|_2^2$

2. Show that

$$f(y) \leq f(x) + \nabla f(x)^\top (y-x) + \frac{M}{2}\|y-x\|^2$$

3. Show that $f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{t}{2}\|\nabla f(x^{(k)})\|^2$

4. Show that

$$f(x^{(k+1)}) \leq v^\# + \nabla f(x^{(k)})^\top (x^{(k)} - x^\#) - \frac{t}{2}\|\nabla f(x^{(k)})\|^2$$

5. Deduce that

$$f(x^{(k+1)}) \leq v^\# + \frac{1}{2t}(\|x^{(k)} - x^\#\|^2 - \|x^{(k+1)} - x^\#\|^2)$$

6. Show that

$$\sum_{i=1}^k f(x^{(i)}) - v^\# \leq \frac{1}{2t}\|x^{(0)} - x^\#\|^2$$

7. Conclude that

$$f(x^{(k)}) - v^\# \leq \frac{1}{2kt}\|x^{(0)} - x^\#\|^2.$$

Answers:

1. We have $\nabla^2 f(x) \preceq MI$, thus $(y-x)^\top (MI - \nabla^2 f(x))(y-x) \geq 0$.

2. Using Taylor remainder theorem we have the existence of $z \in [x, y]$ such that

$$f(y) = f(x) + \nabla f(x)^\top (y-x) + \frac{1}{2}(y-x)^\top \nabla^2 f(z)(y-x)$$

3. We obtain

$$f(x^{(k+1)}) \leq f(x^{(k)}) - (1 - \frac{Mt}{2})\|\nabla f(x^{(k)})\|_2^2$$

and with the condition on t we have $1 - Mt/2 \geq 1/2$.

4. We use the convexity inequality to get $f(x^{(k)}) \leq f(x^\#) + \nabla f(x^{(k)})^\top (x^{(k)} - x^\#)$ and the result of the previous question.

5. We have

$$\begin{aligned} & \frac{1}{2t}(\|x^{(k)} - x^\#\|^2 - \|x^{(k+1)} - x^\#\|^2) \\ &= \frac{1}{2t}(\|x^{(k)} - x^\#\|^2 - \|x^{(k)} - x^\# - t\nabla f(x^{(k)})\|^2) \\ &= \nabla f(x^{(k)})^\top (x^{(k)} - x^\#) - \frac{t}{2}\|\nabla f(x^{(k)})\|^2 \end{aligned}$$

Thus,

$$\begin{aligned} f(x^{(k+1)}) &\leq v^\# + \nabla f(x^{(k)})^\top (x^{(k)} - x^\#) - \frac{t}{2}\|\nabla f(x^{(k)})\|^2 \\ &= v^\# + \frac{1}{2t}(\|x^{(k)} - x^\#\|^2 - \|x^{(k+1)} - x^\#\|^2) \end{aligned}$$

6. Sum the previous inequality

7. As $f(x^{(i)}) - v^\#$ is non-increasing we have that the last term of the sum is lower than the mean of the sum.

Exercise 4 (Kelley's convergence). We are going to prove that, if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, and X a non-empty polytope (bounded polyhedron) then Kelley's cutting plane algorithm is converging.

Consider $x_1 \in X$. We consider a sequence of points $(x^{(k)})_{k \in \mathbb{N}}$ such that $x^{(k+1)}$ is an optimal solution to

$$\begin{aligned} (\mathcal{P}^{(k)}) \quad & \underline{v}^{(k+1)} = \min_{x \in X} \quad z \\ & \text{s.t.} \quad f(x^{(\kappa)}) + \langle g^{(\kappa)}, x - x^{(\kappa)} \rangle \leq z \quad \forall \kappa \in [k] \end{aligned}$$

where $g^{(k)} \in \partial f(x^{(k)})$.

Denote $v = \min_{x \in X} f(x)$.

1. Show that v exists and is finite, and that there exists a sequence $x^{(k)}$.
2. Show that there exists L such that, for all k_1 and k_2 , we have $\|f(x^{(k_1)}) - f(x^{(k_2)})\| \leq L\|x^{(k_1)} - x^{(k_2)}\|$, and $\|g^{(k)}\| \leq L$.
3. Let $K_\varepsilon = \{k \in \mathbb{N} \mid f(x^{(k)}) > v + \varepsilon\}$ be the set of index such that $x^{(k)}$ is not an ε -optimal solution. Show that $f(x_k) \rightarrow v$ if and only if K_ε is finite for all $\varepsilon > 0$.
4. Consider $k_1, k_2 \in K_\varepsilon$, such that $k_2 > k_1$. Show that
$$f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle \leq \underline{v}^{(k_2)} \leq v$$
5. Show that $\varepsilon + f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle < f(x^{(k_2)})$
6. Show that $\varepsilon < 2L\|x^{(k_2)} - x^{(k_1)}\|$.
7. Prove that $f(x^{(k)}) \rightarrow v$.
8. (Optional - hard) Find a complexity bound for the method (that is a number of iteration N_ε after which you are sure to have obtained a ε -optimal solution).

4. By subgradient inequality $f(y) \geq f(x^{(k)}) + \langle g^{(k)}, y - x^{(k)} \rangle$. Thus, for all k , $v \geq \underline{v}^{(k)}$. Further, note that $\underline{v}^{(k)} = f(x^{(k)})$, hence using $k = k_1$, and $y = x^{(k_2)}$ we get

$$f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle \leq \underline{v}^{(k_2)} \leq v.$$

5. As $k_2 \in K_\varepsilon$, we have $f(x^{(k_2)}) = \underline{v}^{(k_2)} > v + \varepsilon \geq f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle + \varepsilon$ by the previous question.

6. We have

$$\varepsilon < |f(x^{(k_2)}) - f(x^{(k_1)})| + |\langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle| \leq 2L\|x^{(k_2)} - x^{(k_1)}\|$$

by Cauchy-Schwartz and question 2.

7. If $f(x^{(k)}) \not\rightarrow v$, then there exists $\varepsilon > 0$ such that $(x^{(k)})_{k \in K_\varepsilon}$ is not finite. As X is compact we can extract a converging subsequence, that is $x^{(\sigma(k))}$ such that $x^{(\sigma(k))} \rightarrow x^*$ and $\sigma(k) \in K_\varepsilon$, which is in contradiction with the result of 6.

Answers:

1. f is finite convex and thus continuous on X which is compact, yielding the existence and finiteness of v .

 f is subdifferentiable, thus we have the existence of $g^{(k)}$, and an optimal solution to $\mathcal{P}^{(k)}$ exists as the solution of a bounded linear program.
2. We have seen that on any compact K included in the domain of a convex function f , f is L -Lipschitz. Here $\text{dom}(f) = \mathbb{R}^n$, so on the compact $K = X + B(0, \varepsilon)$ f is L -Lipschitz, and on X any subgradient g is of norm lower than L .
3. $f(x_k) \rightarrow v$ iff $\forall \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N}, k \geq K \implies f(x_k) \leq v + \varepsilon$. Hence $K_\varepsilon \subset [N_\varepsilon]$.